

Module 02: Exploratory Data Analysis (EDA)

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Topic: Data Visualization & Statistical Inference for ML

1 Introduction: The Detective Work of Machine Learning

Exploratory Data Analysis (EDA) is the rigorous investigative process of understanding your data before attempting to model it. It is the critical first step in the Machine Learning pipeline, governed by the immutable principle: ****Garbage In, Garbage Out****.

If you feed a Neural Network data that is full of outliers, skewed distributions, or hidden biases, it will confidently produce garbage predictions. EDA is not just about making charts; it is about:

- **Understanding Structure:** Is the data normal? Is it multimodal?
- **Cleaning:** Identifying missing values and outliers.
- **Relationship Hunting:** Finding which features actually impact the target.
- **Hypothesis Testing:** Validating your assumptions about the domain.

2 1. Univariate Analysis: Shapes and Centrality

Univariate analysis focusses on a single variable at a time (x_i). Our goal is to describe its **Central Tendency** (Where is the data?) and its **Dispersion** (How spread out is it?).

2.1 Distributions and the Location of Mean, Median, Mode

The "shape" of the data is determined by its probability distribution. The relative positions of the three measures of central tendency reveal the direction of the skew.

- **Normal Distribution (No Skew):** The classic bell curve. The data is symmetrical.

$$\text{Mean} \approx \text{Median} \approx \text{Mode}$$

- **Right-Skewed (Positive Skew):** The tail extends to the **right** (towards positive infinity). Common in financial data (Salaries, House Prices). The outliers pull the Mean upwards.

$$\text{Mode} < \text{Median} < \text{Mean}$$

- **Left-Skewed (Negative Skew):** The tail extends to the **left**. Common in data with a hard upper limit (e.g., Test scores where many get 100%, or Age at Death). The outliers pull the Mean downwards.

$$\text{Mean} < \text{Median} < \text{Mode}$$

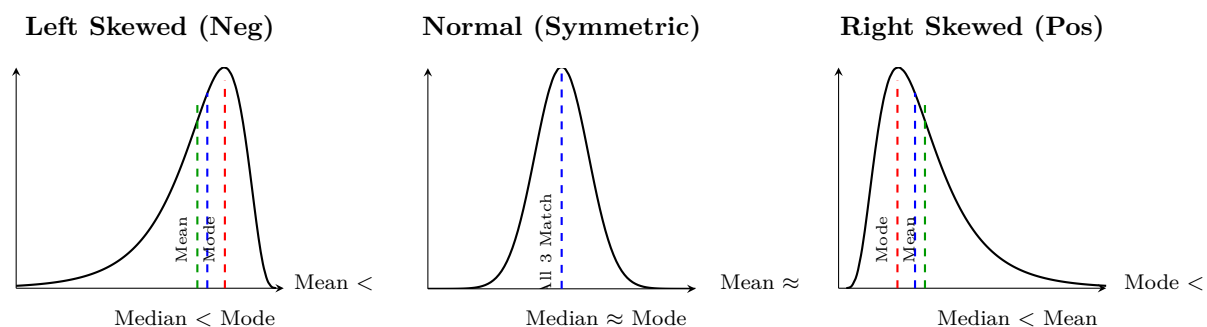


Figure 1: Visualizing Central Tendency across different distributions. In skewed data, the Mean chases the tail, while the Mode stays at the peak.

2.2 Measures of Dispersion

Knowing the center is not enough. We must know the spread.

The Mathematical Definition of Spread

Variance (σ^2): The average of the squared differences from the Mean.

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Standard Deviation (σ): The square root of the variance. This brings the unit back to the original scale (e.g., from "dollars squared" back to "dollars").

$$\sigma = \sqrt{\sigma^2}$$

Why this matters:

- **Low σ :** Data points tend to be very close to the mean. The mean is a "reliable" predictor.
- **High σ :** Data points are spread out over a large range of values. The mean is a poor representation of the "typical" element.

3 2. Bivariate Analysis: Scatter Plots & Correlation

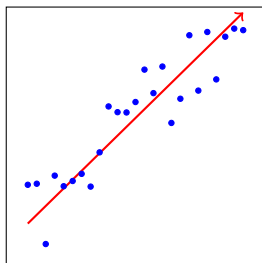
Bivariate analysis explains how Feature X affects Target Y . This is the hunt for **Predictive Power**.

3.1 Understanding Scatter Plots

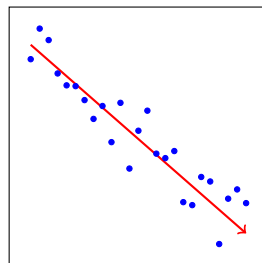
A scatter plot reveals the **Direction**, **Shape**, and **Strength** of a relationship.

1. **Positive Correlation:** As X increases, Y increases. (Slope is Up). Example: Years of Experience vs. Salary.
2. **Negative Correlation:** As X increases, Y decreases. (Slope is Down). Example: Car Age vs. Price.
3. **No Correlation:** X moves randomly regardless of Y . (Cloud of noise). Example: Shoe Size vs. IQ.

Positive Correlation



Negative Correlation



No Correlation

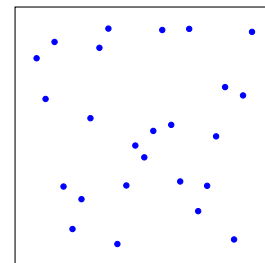


Figure 2: Visualizing Relationships. **Left:** Strong Positive ($r \approx 0.9$). **Center:** Strong Negative ($r \approx -0.9$). **Right:** Zero Correlation ($r \approx 0$).

3.2 Correlation Heatmaps

The Scatter Plot shows the *shape*, but the ****Correlation Matrix**** gives us the exact *number*, known as the ****Pearson Correlation Coefficient (r)****.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

We visualize this matrix as a ****Heatmap****. This is the single most important plot for feature selection.

How to read a Heatmap:

- ****The Diagonal is always 1.0:**** A variable is always perfectly correlated with itself.
- ****Red/Hot Colors (> 0):**** Positive Correlation. Darker red = Stronger relationship.
- ****Blue/Cold Colors (< 0):**** Negative Correlation. Darker blue = Stronger inverse relationship.
- ****Near Zero (White/Grey):**** No linear relationship.

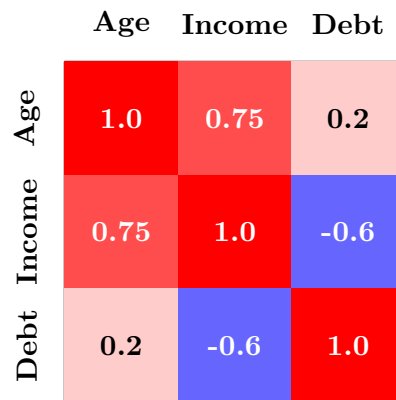


Figure 3: Conceptual Heatmap. Note how Income vs Debt is Blue (Negative): In this hypothetical dataset, Higher income correlates with lower debt ratios.

Python Implementation for Heatmap:

```

1 import seaborn as sns
2 import matplotlib.pyplot as plt
3
4 # 1. Calculate Correlation Matrix
5 corr_matrix = df.corr()
6
7 # 2. Plot Heatmap
8 plt.figure(figsize=(10, 8))
9 sns.heatmap(corr_matrix,
10             annot=True,          # Show numbers
11             cmap='coolwarm',     # Red=Pos, Blue=Neg
12             vmin=-1, vmax=1,    # Scale from -1 to 1
13             square=True)
14 plt.title('Correlation Heatmap')
15 plt.show()

```

4 3. Outlier Detection: The Box Plot

An outlier is a data point that differs significantly from other observations. They can be caused by measurement errors (noise) or can represent real, extreme phenomena (signal).

4.1 The IQR Method

The Box Plot is the standard tool for detecting outliers. It relies on the Median and Quartiles, making it **robust** against the very outliers it detects.

- **IQR (Interquartile Range):** $Q3 - Q1$. This is the height of the box. It contains the middle 50% of the data.
- **Whiskers:** These extend $1.5 \times IQR$ from the box edges.
- **Outliers:** Anything strictly outside the whiskers.

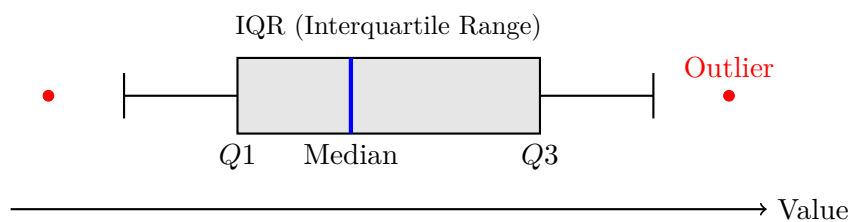


Figure 4: Box Plot Structure. The box plot is essentially a "Top-Down" view of a distribution, specifically designed to flag anomalies.

5 4. Categorical Data Analysis

Not all data is numerical. We often deal with categories (Gender, City, Product Type).

5.1 Cardinality and Class Imbalance

Cardinality refers to the number of unique categories in a feature.

- **Low Cardinality:** "Gender" (Male/Female). Easy to encode.
- **High Cardinality:** "Zip Code" (Thousands of unique values). Difficult to encode.

Class Imbalance occurs when one category dominates the others.

```
1 # Check for imbalance
2 print(df['Target_Class'].value_counts(normalize=True))
3
4 # Output:
5 # No Fraud    0.99
6 # Fraud       0.01
```

If you train a model on this, it will simply learn to predict "No Fraud" every time and achieve 99% accuracy, while failing completely at its actual job. EDA helps us detect this so we can apply resampling techniques later.

6 5. Data Quality and Missing Values

Before any modeling, we must audit the data quality.

6.1 Types of Missing Data

1. **MCAR (Missing Completely At Random)**: No pattern. Safe to drop or impute.
2. **MNAR (Missing Not At Random)**: The missingness is a signal. Example: People with very high debts might leave the "Debt" field blank on a survey. If we fill this with the "Average Debt", we are introducing massive bias.

Golden Rule of Imputation

Never impute based on the whole dataset! You must split your data into **Train** and **Test** sets first, calculate the Mean/Median on the **Train** set, and then apply that value to the Test set. This prevents Data Leakage.

7 Conclusion

EDA is an iterative loop, not a linear path. You visualize, clean, visualize again, and refine. By the end of this module, you should be able to:

- Identify if your data is skewed (and thus needs Log Transformation).
- Detect outliers (and decide whether to clip or remove them).
- Spot strong correlations (to identify your best features).
- Recognize class imbalance (to prepare for resampling strategies).